

Plan

- Row Echelon Form
- Elementary Row Operations
- Gaussian Elimination Method
- Reduced Row Echelon Form
- Gauss-Jordan Elimination Method
- Rank

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- The following matrices are in row echelon form:

$$\begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & -1 & -1 \\ 0 & 1 & 5 \\ 0 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & -1 & 3 & 2 \\ 0 & 0 & 5 & 10 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

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- If a matrix A is in row echelon form, then in each column of A containing a leading entry, the entries below that leading entry are zero.

- The following matrices are **not** in row echelon form:

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- 3 Add a multiple of a row R_j to another row R_i ($R_i \rightarrow R_i + cR_j$).

Example

Transform the following matrix to row echelon form

$$\begin{bmatrix} 0 & 2 & 3 & 8 \\ 2 & 3 & 1 & 5 \\ 1 & -1 & -2 & -5 \end{bmatrix}.$$

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Row Equivalent Matrices: Matrices A and B are said to be **row equivalent** if there is a finite sequence of elementary row operations that converts A into B or B into A .

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*Matrices A and B are row equivalent **if and only if (iff)** they can be reduced to the same row echelon form.*

Linear System of Equations:

A linear system of m equations in n unknowns $x_1, x_2, \dots, x_n$ is a set of equations of the form

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n &= b_2 \\ \vdots & \quad \quad \quad \vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n &= b_m, \end{aligned} \tag{1}$$

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where $a_{ij}, b_i \in \mathbb{C}$ for each $1 \leq i \leq m$ and $1 \leq j \leq n$. Letting

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \quad \text{and} \quad \mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix},$$

we can represent the above system of equations as $A\mathbf{x} = \mathbf{b}$.

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$$[A | \mathbf{b}] = \left[\begin{array}{cccc|c} a_{11} & a_{12} & \dots & a_{1n} & b_1 \\ a_{21} & a_{22} & \dots & a_{2n} & b_2 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} & b_m \end{array} \right].$$

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- The vertical bar is used in the augmented matrix $[A | \mathbf{b}]$ only to distinguish the column vector $\mathbf{b}$ from the coefficient matrix A .
- If $\mathbf{b} \neq \mathbf{0}$ then $A\mathbf{x} = \mathbf{b}$ is called a **non-homogeneous** system of equations.

- If $\mathbf{b} = \mathbf{0} = [0, 0, \dots, 0]^t$, i.e., if $b_1 = b_2 = \dots = b_m = 0$, the system $A\mathbf{x} = \mathbf{0}$ is called a **homogeneous** system of equations.

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- A **solution** of the linear system $A\mathbf{x} = \mathbf{b}$ is a column vector $\mathbf{y} = [y_1, y_2, \dots, y_n]^t$ such that the linear system (1) is satisfied by substituting y_i in place of x_i . That is, $A\mathbf{y} = \mathbf{b}$ holds true.

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Result

Let $C\mathbf{x} = d$ be the linear system obtained from the linear system $A\mathbf{x} = b$ by a single elementary row operation. Then the linear systems $A\mathbf{x} = b$ and $C\mathbf{x} = d$ have the same set of solutions.

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Gaussian Elimination Method: Use the following steps to solve a system of equations $A\mathbf{x} = \mathbf{b}$.

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- 3 Use **back substitution** to solve the equivalent system that corresponds to the row echelon form.

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Use Gaussian Elimination method to solve the system:

(a) $y + z = 1, \quad x + y + z = 2, \quad x + 2y + 2z = 3$

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- If $A\mathbf{x} = \mathbf{b}$ has at least one solution then it is called a **consistent** system. Else, it is called an **inconsistent** system.

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Then the statement $P(n)$ is true for all positive integers.

Proof: RREF is unique (Using Induction)

We will apply induction on number of columns.

Base Case: Consider a matrix with one column i.e. $A_{m \times 1}$.
Observe that its RREF is unique. What are the possibilities?
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- Assume A has two RREF say B and C . Note that because of induction hypothesis the submatrices B_1 and C_1 formed by the first $k - 1$ columns of B and C , respectively, are identical.

Proof: RREF is unique (Using Induction)

We will apply induction on number of columns.

Base Case: Consider a matrix with one column i.e. $A_{m \times 1}$. Observe that its RREF is unique. What are the possibilities? $\mathbf{0}_{m \times 1}$ and e_1 . Can you transform $\mathbf{0}_{m \times 1}$ into e_1 ?

Induction Hypothesis: Let any matrix having $k - 1$ columns have a unique RREF.

Inductive Case:

- Consider a matrix $A_{m \times k}$.
- We need to show that A also has unique RREF.
- Assume A has two RREF say B and C . Note that because of induction hypothesis the submatrices B_1 and C_1 formed by the first $k - 1$ columns of B and C , respectively, are identical. In particular, they have same number of nonzero rows.

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Result

Matrices A and B are row equivalent iff $RREF(A) = RREF(B)$.

Method of transforming a given matrix to reduced row echelon form: Let A be an $m \times n$ matrix. The following step by step method is used to obtain the reduced row echelon form of A .

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- (2) Forget the first row and first i columns. Start with the lower $(m - 1) \times (n - i)$ sub matrix of the matrix obtained in the first step and proceed as in **Step 1**.

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- (3) Repeat the above steps until we get a row echelon form.
- (4) Now use the leading term in each of the leading column to make (by elementary row operations) all other entries in that column equal to zero. Use this step starting from the rightmost leading column.
- (5) Scale the leading entries of the matrix obtained in the previous step, by multiplying the rows by suitable constants, to make all the leading entries equal to 1, ending with the unique reduced row echelon form of A .

Gauss-Jordan Elimination Method: Use the following steps to solve a system of equations $A\mathbf{x} = \mathbf{b}$.

- 1 Write the augmented matrix $[A \mid \mathbf{b}]$.

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Example

Solve the system

$w - x - y + 2z = 1$, $2w - 2x - y + 3z = 3$, $-w + x - y = -3$
using Gauss-Jordan elimination method.

Example

Solve the following systems using Gauss-Jordan elimination method:

(a) $2y + 3z = 8$, $2x + 3y + z = 5$, $x - y - 2z = -5$

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Result

Let $Ax = \mathbf{b}$ be a system of equations with n variables. Then number of free variables is equal to $n - \text{rank}(A)$.

Result

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